

1.4.27

EE25BTECH11011 - Navya

Question:

If $\mathbf{a}$ and $\mathbf{b}$ are the position vectors of A and B respectively, find the position vector of a point C in BA produced such that $BC = 1.5 BA$

Solution:

Given the position vectors of A and B are $\mathbf{a}$ and $\mathbf{b}$

Let the position vector of point C be $\mathbf{c}$

The vector BC is given by:

$$\mathbf{BC} = 1.5\mathbf{BA} \quad (1)$$

$$(\mathbf{c} - \mathbf{b}) = 1.5(\mathbf{b} - \mathbf{a}) \quad (2)$$

we have

$$\mathbf{AC} = \mathbf{BC} - \mathbf{BA} \quad (3)$$

$$(\mathbf{c} - \mathbf{a}) = (\mathbf{c} - \mathbf{b}) - (\mathbf{a} - \mathbf{b}) \quad (4)$$

$$(\mathbf{c} - \mathbf{a}) = (1.5 - 1)(\mathbf{a} - \mathbf{b}) \quad (5)$$

$$(\mathbf{c} - \mathbf{a}) = 0.5(\mathbf{a} - \mathbf{b}) \quad (6)$$

Thus,

$$(\mathbf{c} - \mathbf{a}) : (\mathbf{b} - \mathbf{c}) = 0.5 : 1.5 = 1 : 3 \quad (7)$$

Therefore, C divides AB externally in the ratio $1 : 3$

Writing the ratio as $k : 1$ we have

$$k = 3 \quad (8)$$

By the section formula (external division), the position vector of C is

$$\mathbf{c} = \frac{k\mathbf{b} - \mathbf{a}}{k - 1}, \quad (k = 3). \quad (9)$$

$$\mathbf{c} = \frac{3\mathbf{b} - \mathbf{a}}{2}. \quad (10)$$

$$\therefore \mathbf{C} = \frac{3}{2}\mathbf{b} - \frac{1}{2}\mathbf{a}. \quad (11)$$

The above equation can be written as:

$$\mathbf{c} = (\mathbf{a} \quad \mathbf{b}) \begin{pmatrix} \frac{3}{2} \\ -\frac{1}{2} \end{pmatrix} \quad (12)$$

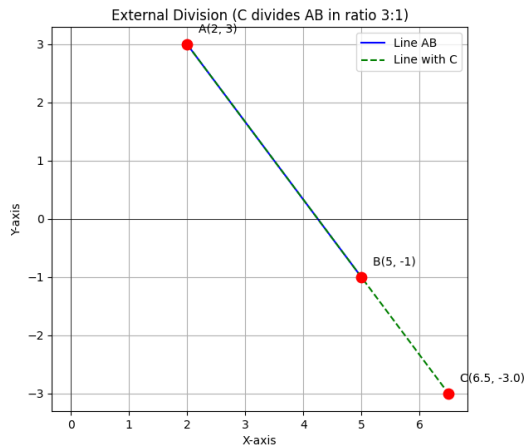


Fig. 0: Point C dividing AB externally in ratio 3:1